Histology

It is recommended to take a lesional skin biopsy for histopathology.

A standardized (buffered) 4% formaldehyde (10% formalin) solution should be used for storage and transport of the biopsy specimen.

A $4–5\,\mathrm{mm}$ lesional punch biopsy is recommended. An intact vesicle or small erosion on inflamed skin may be completely excised. In the case of larger blisters, which are rare, it is advised to include a small portion of perilesional skin (approximately $\frac{1}{4}$ of the biopsy) to prevent detachment of the blister roof during processing.

The histopathological hallmarks of dermatitis herpetiformis (DH) include accumulation of neutrophils forming neutrophilic microabscesses in the dermal papillae. Dermal papillary oedema is commonly observed, and subepidermal splitting is present in fully developed lesions. The blister cavity typically contains fibrin, polymorphonuclear leukocytes (predominantly neutrophils), and nuclear dust. Eosinophils may be present in variable numbers. A similar inflammatory infiltrate can also be found in the upper dermis. In older lesions, nonspecific changes such as lymphocytic infiltration, fibrosis, and ectatic capillaries may be seen.

However, histopathological findings alone cannot differentiate DH from other autoimmune bullous diseases such as linear IgA disease, bullous pemphigoid, anti-p200/laminin $\gamma 1$ pemphigoid, or the inflammatory variant of epidermolysis bullosa acquisita.

Direct Immunofluorescence Microscopy

Direct immunofluorescence (DIF) microscopy is the gold-standard laboratory procedure for diagnosing DH.

The optimal biopsy site for DIF is non-lesional skin, preferably from a perilesional area or a predilection site such as the extensor surfaces of the elbows, knees, or buttocks.

DIF can be performed using cryosections examined under a short-arc mercury lamp, blue light-emitting diode (LED) microscopy, or laser scanning confocal microscopy. The characteristic findings in DH are microgranular or microgranular-fibrillar IgA deposits located at the tips of the dermal papillae and along the dermo-epidermal junction (DEJ). These papillary IgA deposits often appear as vertical linear structures resembling "falling snow." With serial sectioning, up to seven different patterns may be observed due to combinations of the three primary deposition patterns.

In approximately 30% of Japanese patients, a fibrillar IgA deposition pattern has been reported. Both $\mathrm{IgA_1}$ and $\mathrm{IgA_2}$ subclasses can form cutaneous deposits in DH, although $\mathrm{IgA_1}$ is the predominant subclass.